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Sustainability information overload: Its effect on customers' greenwashing perceptions, perceived value, and behavioral intentions

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ABSTRACT

Little is known about the effect of perceived sustainability information overload on customers' perceptions of greenwashing, perceived value, and behavioral intentions, nor how this effect is contingent on the customers' expectations of sustainability practices. To address the impact of information overload, we integrate the Stimuli-Organism-Response (S-O-R) model with the media richness theory, and the expectation-confirmation theory. An online survey measured 808 British and German travelers' expectations of hotels' sustainability practices and exposed them to a sustainability scorecard. The results show that perceived information overload increases perceived greenwashing and has a negative impact (via greenwashing) on ethical-expressive and altruistic values. This is a challenge for hotel managers, as perceived value positively affects self-reported booking intentions and electronic Word of Mouth (eWOM) recommendations. Furthermore, customer expectations of sustainability practices moderate i) the positive relationship between information overload and greenwashing, and ii) the relationship between greenwashing and perceived value. Higher customer expectations of sustainability practices strengthen the effect of perceived information overload on perceived greenwashing. However, unexpectedly, the higher expectations of sustainability practices weaken the impact of perceived greenwashing on ethical-expressive and altruistic values, suggesting that customers with high sustainability expectations may be more forgiving of a certain amount of greenwashing than others.

1. Introduction

Sustainability communication is often used to influence consumer purchases by highlighting product features that satisfy their needs and meet socially expected sustainability standards (Golob et al., 2023; Tölkes, 2018). Greenwashing refers to the misuse of sustainability communication to give a false impression of a product's sustainability features or benefits, to deliberately mislead customers. Greenwashing in sustainable communication is gaining increasing attention among tourism and hospitality scholars (Chen et al., 2019; Majeed & Kim, 2023; Papagiannakis et al., 2024). There is a fine line between what is considered greenwashing and what is not, and customers' perceptions of greenwashing change over time based on their knowledge and expectations. While sustainability claims help customers find hotels that are operating more sustainably (Arzoumanidis et al., 2022; Huang et al., 2024; Kim et al., 2021), businesses that make such claims must understand how to communicate sustainability information in ways that avoid the risk of miscommunication. Prior research suggests that customers' cognitive elaborations of sustainability information can result in more eco-friendly behaviors (Acuti et al., 2022). Many firms assume that more sustainability communication is good (Gonzalez-Arcos et al., 2021; Szabo & Webster, 2021; Yan et al., 2024). However, sustainability communication can be too frequent, ambiguous, and undifferentiated, leading to customer confusion (Torma & Thøgersen, 2024) and perceived greenwashing (Peng & Chen, 2019).

Hence, we examine sustainability communication using the Stimulus-Organism-Response (S-O-R) model and the media richness theory. Previous scholars posit that perceived information overload acts as a stimulus (Hu et al., 2023; Lin et al., 2020). Such information overload affects behavior, mediated by an organism's internal processes, in this case, via customers' perceptions, feelings, and thoughts (Lin et al., 2020). Media richness theory suggests that communication uses various cues to help receivers reduce ambiguity and uncertainty by providing social, emotional, or task-related context (Otondo et al., 2008).

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Paradoxically, the richness of information can complicate customers' decision-making processes through information overload (Longo et al., 2019)—customers may be confused due to the overload or fail to understand the information (Yang et al., 2021). To communicate their sustainability initiatives, hotels can apply green marketing strategies to appeal to sustainability-conscious customers (Szabo & Webster, 2021). However, some companies exaggerate their sustainability claims (Chen et al., 2019). Focusing on hospitality, our first contribution is to analyze how information overload of hotels' sustainability initiatives might influence perceived greenwashing.

We further examine perceived value as an additional organism-related factor, as per the S-O-R model (Hu et al., 2023). While studies show that greenwashing has a negative impact on sustainability value, there is a call for further research into its impact on other types of value (see Szabo & Webster, 2021). To that end, using three dimensions of perceived value (social, ethical-expressive, and altruistic), we examine how perceived greenwashing influences perceived value and behavioral intentions (booking intentions and electronic word-of-mouth, eWOM, recommendations). Relying on the expectation-confirmation theory (Olkkonen & Luoma-Aho, 2015), we further contribute to the sustainability literature by examining the moderating effect of customers' sustainability expectations. Previous research suggests that customers' sustainability knowledge and expectations may determine the impact of greenwashing (Chen et al., 2019; Kim et al., 2023; Majeed & Kim, 2023). However, there is scant research on how customer expectations of hotels' sustainability initiatives moderate the effect of perceived information overload on greenwashing.

The present research examines how perceived overload of sustainability information affects customers' perceptions of greenwashing in the online booking context. Specifically, we examine i) the effect of perceived information overload on greenwashing perceptions, and ii) how greenwashing perceptions negatively influence perceived value and, in turn, customers' intentions to book and give eWOM recommendations. Also, we examine iii) the moderating role of customers' expectations about hotels' sustainability initiatives on the relationship between perceived information overload and perceived greenwashing, and iv) how customers' expectations about hotels' sustainability initiatives moderate the relationship between perceived greenwashing and perceived value.

2. Theoretical foundation and hypotheses

In 1974, Mehrabian and Russell (1974) introduced the S-O-R model, showing that exposure to a stimulus (S) creates internal states in an organism (O) that lead to responses (R). The internal states mediate the relationship between the external stimuli and the organism's responses. This S-O-R model has been verified across different studies of environmental communication in hotels (Hameed et al., 2022; Tan, 2023) and at destinations (Su & Swanson, 2017). Hameed et al. (2022) analyzed the effect of green practices on customers' green word-of-mouth intentions for hotels. Their study used green practices and green images as stimuli; green satisfaction and trust as organisms; and green word-of-mouth as a response. Tan (2023) examined the relationships among monetary cost (stimulus), fear of perceived greenwash, attitude, and perceived behavioral control (organism), and green hotel patronage intention (response). In the context of destinations, Su and Swanson (2017) found that the perceived responsible actions of a destination (stimuli) trigger consumption emotions and destination identification (organism), leading to environmentally responsible behaviors (response).

Previous studies considered sustainability stimuli to be positive. Conversely, we aim to study what happens when sustainability communication is perceived negatively. We complement the S-O-R model with the media richness theory (Otondo et al., 2008) and the expectation-confirmation theory (Olkkonen & Luoma-Aho, 2015). This allows us to examine the effect of perceived sustainability information overload (the stimulus) on two internal psychological reactions, perceived greenwashing and perceived value (the organisms), and, how these generate two specific behavioral intentions, booking intention and eWOM (the responses). Additionally, we analyze the moderator effect of customer expectations of sustainability initiatives on the relationship between information overload and internal states.

2.1. Sustainability information overload leads to perceptions of greenwashing

Based on the S-O-R model, information overload represents the stimulus that influences organism-related factors (Hu et al., 2023). The media richness theory (Otondo et al., 2008) suggests that the right amount of information in a stimulus can reduce uncertainty, which positively influences the communication outcome. The purpose and the content of a message determine the appropriate amount. If the medium's capacity is low, some information cannot be transmitted, and communication will be less effective. Richer media convey more cues, but too many can lead to information overload.

Information overload "is a state in which a decision maker faces information comprising the accumulation of individual informational cues of differing size and complexity that inhibit the decision maker's ability to optimally determine the best possible decision" (Roetzel, 2019, p. 484). Information overload occurs when information processing requirements exceed one's processing abilities (Matthes et al., 2020), with information becoming a hindrance despite its relevance and potential usefulness (Bawden & Robinson, 2020). Information overload can also occur when information is unfamiliar, complex, or presented faster than the receiver can process (Otondo et al., 2008).

Online booking platforms can play an important role in streamlining and communicating sustainability information at the point of purchase (Arzoumanidis et al., 2022; Huang et al., 2024), yet there is little empirical evidence to back up this argument. Information overload is more likely in virtual (vs. traditional) retail environments due to i) the overload of customers' cognitive abilities coupled with ii) the reduced degree of human interaction to interpret such information (Li, 2017). For example, increasing the quantity of environmental information on food labels has been shown to decrease customer recall and comprehension accuracy, leading to cognitive overload and confusion (Bogliacino et al., 2023). A basic eco-ranking (vs. rankings with multiple indicators) is more successful in reducing uncertainty and increasing sustainable choices (Weber, 2021).

Based on the S-O-R model, perceived value has been examined as part of the organism-related factors (Hu et al., 2023). Customers may feel that they are exposed to information overload when the information is not substantiated by trustworthy claims (Gupta et al., 2019). Perceived lack of relevance of sustainability stimuli may increase customers' skepticism about the company's motives. Greenwashing is defined as misleading customers about a company's environmental practices or the environmental benefits of a product or service (Parguel et al., 2011). Guests often believe hotels overstate their sustainability claims (Gupta et al., 2019; Peng & Chen, 2019; Rahman et al., 2015). As suggested by Papagiannakis et al. (2024:1), "customers have difficulty distinguishing whether communicated CSR [Corporate Social Responsibility] actions (e.g., towel reuse) represent window-dressing for the true aim, which is the reduction of operating costs". Therefore, customers might infer that "too much" information about sustainability practices is an attempt to mislead them. We thus put forth the following prediction:

H1. Perceived information overload has a positive impact on customers' perceived greenwashing.

2.2. Perception of greenwashing negatively affects perceived social, ethical-expressive, and altruistic values

Customers who perceive that a company exaggerates its

sustainability claims (greenwashing) distrust that company's communication, and by extension, the channel that communicates it (Chen & Chang, 2013; Chen et al., 2019). Customers are increasingly sensitized to greenwashing and this has a negative impact on their attitudes towards sustainability marketing practices more generally (Chen & Chang, 2013; Parguel et al., 2011). In turn, this skepticism reduces the perceived value of being sustainable (Chen et al., 2022; Szabo & Webster, 2021).

The perceived value of sustainability practices in hotels is conceptualized as a multi-dimensional construct, encompassing not only consumers' comparative assessments of quality and price (i.e., functional value) (Zeithaml, 1988), but also their cognitive appraisal of the extent to which these hotels facilitate identification with aspirational social groups (i.e., social value) and elicit positive affective responses (i.e., emotional value) (Ahn & Kwon, 2020; Teng et al., 2018). Peloza and Shang (2011) suggest that Holbrook's (2006) typology of customer value is particularly salient in the context of sustainability. Several tourism studies have used this typology to demonstrate that a positive perception of hotels' sustainability practices positively influences perceived value (e.g., Marchoo et al., 2014; Mohammed & Al-Swidi, 2019; Teng et al., 2018). Following Holbrook (2006) and Peloza and Shang (2011), we argue that there are three dimensions of perceived value in sustainability: social, ethical-expressive, and altruistic. *Social value* refers to making a good impression and gaining approval from others, while *altruistic value* refers to the coherence of an individual's ethical values (Ahn & Kwon, 2020). Marchoo et al. (2024) identified *ethical-expressive value* as travelers' responses to tourism accreditations and codes of ethics, which result from the moral values gained when travelers enhance their own images/self-concepts as responsible tourists. Conversely, customers respond negatively to *self-serving* sustainability practices (Nyilasy et al., 2014) as these negatively impact on their perceived value from the firm (Szabo & Webster, 2021). We extend the literature by examining the effects of greenwashing on the three dimensions of perceived value. Consequently, we put forth the following prediction:

H2. Perceived greenwashing will have a negative impact on customers' perceptions of (a) social value, (b) ethical-expressive value, and (c) altruistic value.

2.3. Perceived value affects booking intention and electronic word-of-mouth recommendations

Based on the S-O-R model, perceived value is a psychological factor that affects consumers' responses (Hu et al., 2023), typically measured as booking intentions (Tan, 2023) and eWOM recommendations (Hameed et al., 2022). Perceived social benefits drive customer attitudes and behavioral intentions, especially in the context of green marketing (Iyer & Kashyap, 2007). Previous research suggests that customers with altruistic values feel good about choosing a green hotel because they are less likely to harm the environment or people (Pickett-Baker & Ozaki, 2008) or because the hotel evokes positive emotions (Tran et al., 2024). Moreover, having an awareness of the positive consequences (versus negative consequences) of sustainability practices is more effective in driving pro-sustainable behavior (Sajid et al., 2024). The messaging of positive consequences helps reduce eco-paralysis (Li et al., 2024). We argue that exposing consumers to a hotel's sustainability practices can induce more sustainable choices. Recent evidence shows that consumption values associated with the social ethical-expressive and altruistic dimensions positively affect customers' intentions to book (Ray et al., 2024) and they are more likely to recommend hotels with trustworthy sustainability practices (D'Acunto et al., 2024). Consequently, we propose:

H3. Perceived value (social, ethical-expressive, and altruistic) positively influences customers' booking intentions.

H4. Perceived value (social, ethical-expressive, and altruistic) positively influences customers' eWOM recommendations.

2.4. Customer expectations of sustainability practices moderate the relationships between information overload and other variables

We now draw on the expectation-confirmation theory to delve deeper into the potentially negative consequences of ill-designed sustainability stimuli. According to the expectation-confirmation theory, customers' sustainability expectations are critical for their decision-making processes (Olkkonen & Luoma-Aho, 2015). Expectations form a psychological contract that corresponds to the nature of an exchange relationship between the customer and the company (Mason & Simmons, 2012). Sustainability expectations refer to the customer's belief that a company will fulfill certain minimum responsibilities regarding what matters most to them (Kim et al., 2023). For instance, customer expectations of sustainability practices refer to the belief that a company implements environmental programs (e.g., energy-saving systems, recycling waste material), adopts social programs (e.g., improves the quality of life in the local community, sponsors sociocultural practices) and invests in economic sustainability practices (e.g., long-term planning, control of production costs). Customers' expectations influence their responses to sustainability practices (Rim et al., 2020), and they serve as standards against which the actual performance is evaluated (Zeithaml et al., 2018).

Prior advertising research shows that issue-related knowledge is an important factor in evaluating the credibility of advertising claims, especially when the claims are ambiguous (Schmuck et al., 2018). Customers' sustainability expectations influence the perceived credibility of company-initiated communication. Customers with higher-than-average sustainability expectations are more critical of sustainability information. Therefore, we suggest that customer expectations moderate the positive relationship between information overload and perceived greenwashing. While we could not find specific examples of this argument, we can draw parallelisms from evidence that when customers perceive that a company claims to be sustainable while, at the same time, providing excessive product packaging (Qayyum et al., 2023). The same could be argued for hedonic hotel experiences that may be perceived as incongruent with pro-sustainability messages (Chen et al., 2019). Customers with high environmental involvement are more inclined to penalize brands that make poor efforts to be environmentally friendly (Schmuck et al., 2018). Similarly, customers with higher-than-average sustainability expectations make more rigid inferences about a brand's motives, and, therefore, may withdraw their support of the brand's sustainability practices (Kim et al., 2023). Customers with higher sustainability expectations are more likely to recognize misleading intentions due to greater engagement in the detailed processing of advertising claims (Johar, 1995). Consequently, we propose that:

H5. Customers' expectations of sustainability practices moderate the positive relationship between information overload and perceived greenwashing.

As hypothesized before, perceived greenwashing negatively influences customers' perceived value. However, based on the literature above, for customers with higher expectations of sustainability practices, the negative effect of greenwashing on perceived value is expected to be stronger. Therefore, we suggest that:

H6. Customers' expectations of hotels' sustainability practices moderate the negative relationship between greenwashing and their perceptions of (a) social value, (b) ethical-expressive value, and (c) altruistic value.

Fig. 1 shows the conceptual model with the three components of the S-O-R model: perceived information overload (Stimulus); perceived greenwashing and perceived value (Organism); and booking intention

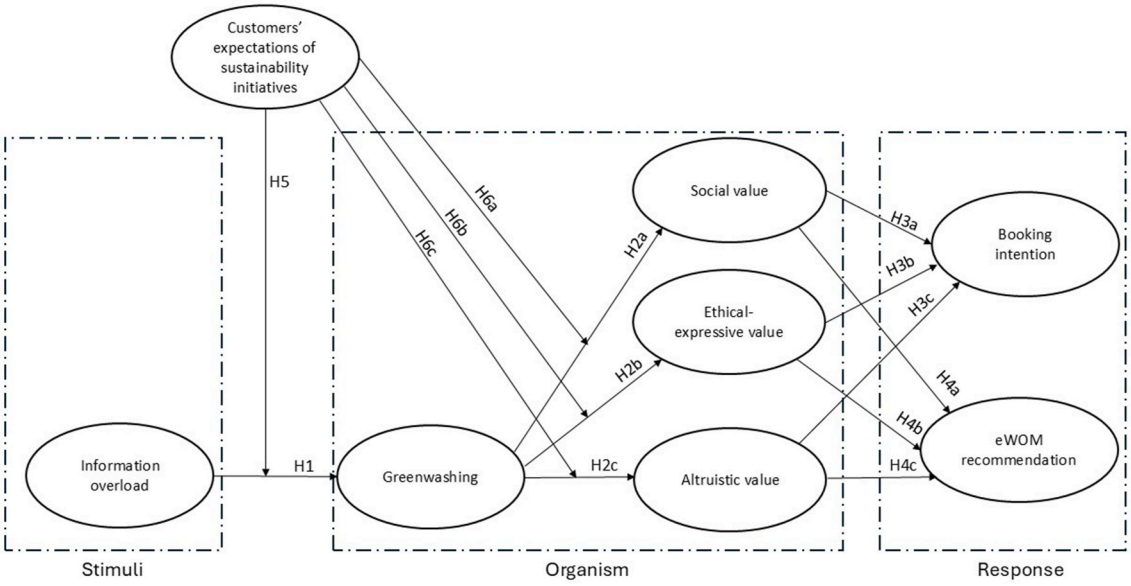


Fig. 1. The effects of sustainability information overload.

and eWOM recommendation (Responses). Additionally, the model integrates the expectation-confirmation theory with the moderating role of customers' expectations of sustainability practices.

3. Materials and methods

We examine how perceived information overload of a hotel's sustainability practices affects customers' perceptions of greenwashing when booking a hotel online. We also investigate the effects of perceived greenwashing on values (social, ethical-expressive, and altruistic) and, in turn, values on behavioral intentions (bookings and recommendations). Furthermore, we analyze how customers' expectations of a hotel's sustainability practices moderate their perceptions of information overload and greenwashing, and the effects of these on perceived value.

3.1. Survey instrument

The stimulus we used to measure perceived sustainability information overload was based on Booking.com's Travel Sustainable measures. At the time of first writing this paper (2023), Booking.com, 2022 collected self-reported sustainability data from over 1.3 million properties. They selected to showcase the highest performing properties (>400,000)—these properties gave extensive details of the practices they claimed to have undertaken (see Fig. 2). To our knowledge, this is the world's largest sustainable hospitality communication effort. In Spring 2024, Booking.com, 2022 removed the Travel Sustainable label, yet each property's sustainability practices remain visible on the property's web page, as shown in this study.

We first measured customers' expectations of hotel sustainability

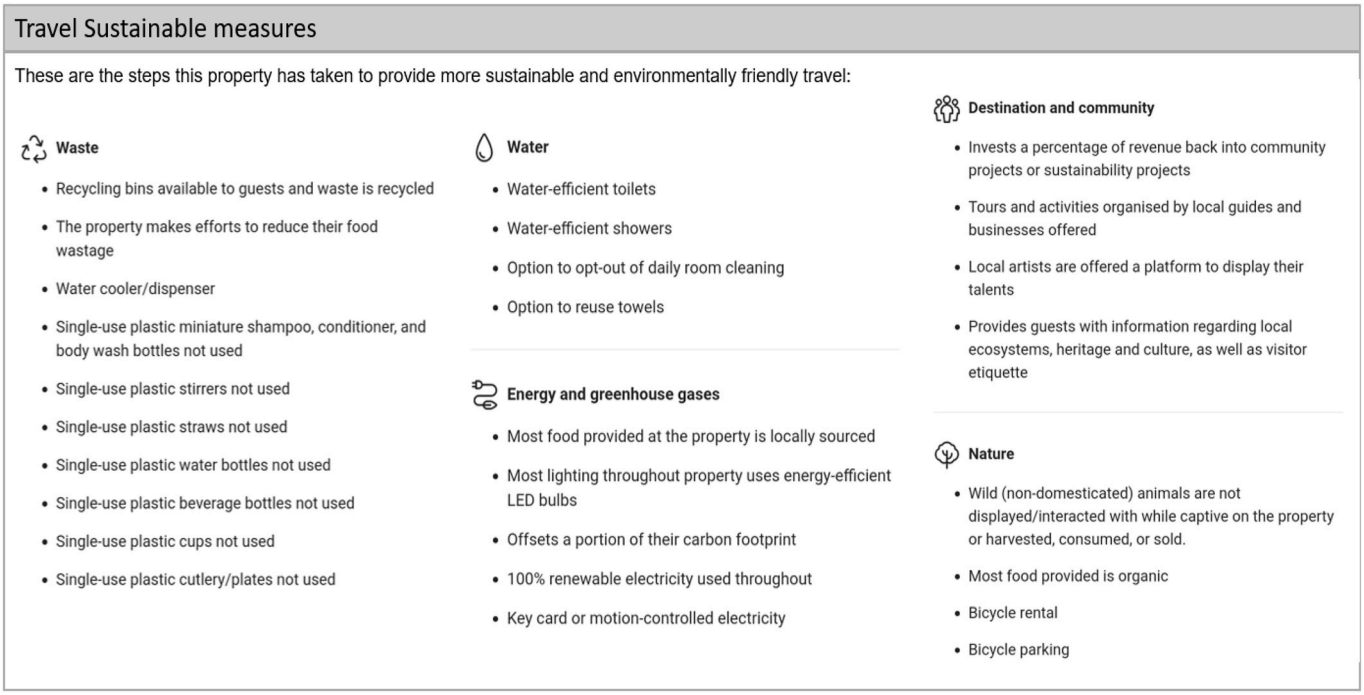


Fig. 2. Research stimuli based on www.booking.com (August 2022).

practices using Alvarado-Herrera et al.'s (2017) scale, modelized as a second-order construct of economic, environmental, and social lower-order dimensions. Next, participants were exposed to Booking.com's Travel Sustainable Scorecard (Fig. 2) and were asked to report their perception of information overload following the scale from Otondo et al. (2008). Perceived greenwashing was captured via Chen et al.'s (2019) scale. We then asked them about the perceived social value (see Ahn & Kwon, 2020), perceived ethical-expressive value (see Marchoo et al., 2014), and perceived altruistic value (see Ahn & Kwon, 2020) associated with a hotel that implements Booking.com's sustainability practices. Finally, we measured their intentions to book (see Chan et al., 2017), their intentions to spread positive eWOM recommendations (see Lee et al., 2010), and their demographic information. The questionnaire was developed in English and, subsequently, translated into German; for both languages, bilingual researchers checked to ensure the linguistic quality of both versions of the questionnaires. Online Appendix 1 shows the measurement scales.

3.2. Data collection

We conducted an online survey using Computer Assisted Web Interview (CAWI) via Cint (www.cint.com), a global consumer network for digital survey-based research. The research population consisted of the two largest outbound tourism markets in Europe and, specifically, individuals aged 18–70 years who had used hotel search platforms (like Booking.com) in the previous 12 months. The data was collected online using quota sampling based on age and gender in September 2022. The final research sample was 808 individuals (from Germany (n = 402) and the United Kingdom (n = 406)). Online Appendix 2 shows the socio-demographic information of the respondents.

3.3. Reliability and validity assessment

The model presented in Fig. 1 was estimated using structural equation modeling based on variances (PLS-SEM) to test our predictions. Steps indicated by Hair et al. (2022) and Becker et al. (2023) were followed. First, the measurement model's reliability, convergent, and discriminant validity had to be confirmed. Second, the results of the structural model were assessed and the hypotheses tested accordingly.

Since a self-reported questionnaire was used to obtain data from the respondents at the same time, to test if common method bias was a methodological problem, the full collinearity test was applied (Kock & Lynn, 2012). The results allowed us to discard this question from being a relevant problem.

3.3.1. Measurement model assessment

To evaluate the measurement and structural model, the model was estimated using the following algorithm settings and SmartPLS 4 (Ringle et al., 2024). As all the latent variables were reflectively related to their scale items, correlation weights between the construct and its indicators were used to compute the construct scores. A path weighting scheme was used to determine the relationships in the structural model.

As PLS-SEM does not assume the data are normally distributed, a bootstrapping of 10,000 subsamples was performed to evaluate the significance of parameters, and we used the percentile method for constructing confidence intervals. The indicator reliability was checked using Cronbach's alpha (Cronbach & Gleser, 1965), composite reliability (Fornell & Larcker, 1981) and rho_A (Dijkstra & Henseler, 2015). Apart from being significant, according to the bootstrap results, a size of 0.70 was required. The average variance extracted (AVE) was calculated to assess convergent validity (Fornell & Larcker, 1981). Values higher than 0.50 indicate that, on average, the construct explains more than half of the variance of the indicators, which is the benchmark followed. The results showed no evidence of reliability or convergent validity problems for either the first- or second-order constructs (see Online Appendix 3).

Two criteria were followed to test discriminant validity (see Online Appendix 4). First, Fornell and Larcker, 1981 compare the square root of the AVE values with the latent variables' correlations, where each square root of the AVE of a latent variable should be higher than each correlation in which that latent variable is implied. Second, the heterotrait-monotrait ratio (HTMT) proposed by Henseler et al. (2015), which uses an upper limit of 0.90. The results showed no evidence of discriminant validity issues.

3.3.2. Evaluation of the structural model

After confirming that the construct measures were reliable and valid, we then evaluated the structural model properties. Collinearity issues were discarded, as the variance inflation factors (VIFs) were lower than 3 (see Online Appendix 5) (Becker et al., 2015; Mason & Perreault, 1991).

The explanatory power of a model relates to its ability to fit the data at hand by quantifying the strength of the association indicated by the PLS path model (Shmueli, 2010; Shmueli & Koppius, 2011). To evaluate the structural model's explanatory power, the coefficient of determination R^2 was used, which represents a measure of in-sample predictive power (Rigdon, 2012). As can be seen in Table 1, for the final dependent variables (booking intention (0.484) and eWOM recommendation (0.574)), our model can explain around 50% of what, according to most rules of thumb, can be considered a moderate explanatory power.

The last step in evaluating the structural model, before focusing on the hypotheses, is to analyze the model's predictive power using the cross-validated predictive ability test (CVPAT). This test was developed by Liengaard et al. (2021) for prediction-oriented model comparison in PLS-SEM and extended by Sharma et al. (2023) for evaluating a model's predictive capabilities.

To demonstrate predictive validity, Sharma et al. (2023) propose that the PLS-SEM predictions should always be significantly better than the naïve indicator-averages (IA) prediction benchmark. We refer to strong predictive validity when the PLS-SEM predictions are significantly better than the linear model (LM) prediction benchmark. As Table 1 shows, the overall test (-0.299; $t = 12.202$ $p < 0.01$) shows that PLS-SEM predictions are significantly better than the indicator averages. This demonstrates prediction validity, but not a strong prediction validity as the overall test (0.016; $t = 2.55$; $p < 0,05$) shows loss is not lower than the LM prediction benchmark.

4. Results

The descriptive results can be found in Online Appendix 1. The fact that 47.9% of the sample tend to agree or totally agree that the current communication from Booking.com is information overload (OVER_02) is concerning. This information is perceived not to fit their needs by 40.7% of the respondents (OVER_03), and not to be easily interpretable by 46.7% of the respondents (OVER_04). Furthermore, 58.85% of the respondents state that hotels that display the type of information currently portrayed by Booking.com overstate or exaggerate how green their operations are (GRW_02), 49% perceive that the adoption of these sustainability claims is vague or seemingly unprovable (GRW_03), and 48.5% state that these hotels leave out, or mask, important information to make their sustainability claims sound better than they really are (GRW_05). The model estimation (see Table 2) demonstrates that perceived information overload increases perceived greenwashing (H1; $\beta = 0.432$; $p < 0.01$), which makes information overload a variable/construct to be carefully managed as it clearly deteriorates the credibility of sustainability practices.

Customers nowadays have high expectations of sustainability practices, ranging from 75% agreeing or strongly agreeing that they expect hotels to recycle (HENV_05), to 53% agreeing or strongly agreeing that hotels are expected to make financial donations to social causes (HSOC_05). Our results demonstrate that the higher these expectations are, the stronger the previously mentioned effect of information

Table 1
Explanatory and predictive relevance.

Factor	Indicators' average (IA)			Indicators' average (LM)			R^2	Adjusted R^2
	(Model-benchmark) loss	t-value		(Model-benchmark) loss	t-value			
Booking intention	−0.300	9.890	**	0.055	3.749	**	0.484	0.482
eWOM recommendation	−0.398	11.361	**	0.041	3.368	**	0.574	0.573
Perceived greenwashing	−0.143	5.677	**	−0.012	1.406		0.229	0.226
Perceived social value	−0.338	8.739	**	0.021	1.554		0.328	0.326
Perceived ethical-expressive value	−0.343	9.520	**	0.004	0.637		0.445	0.443
Perceived altruistic value	−0.374	9.549	**	0.030	2.310	*	0.414	0.411
Overall	**−0.299**	**12.202**	**	**0.016**	**2.550**	*		

**p < 0.01; *p < 0.05.

Table 2
Hypotheses testing.

Hypotheses		Stand. beta		t-value (bootstrapping)	95% CI	97.50%
					2.50%	
H1	Information overload - > Greenwashing	0.432	**	13.300	0.362	0.491
H2a	Greenwashing - > Social value	−0.041		1.255	−0.106	0.024
H2b	Greenwashing - > Ethical value	−0.087	**	2.815	−0.146	−0.026
H2c	Greenwashing - > Altruistic value	−0.113	**	3.750	−0.170	−0.052
H3a	Perceived social value - > Booking intention	0.128	**	3.530	0.058	0.200
H3b	Perceived ethical value - > Booking intention	0.268	**	5.583	0.169	0.357
H3c	Perceived altruistic value - > Booking intention	0.363	**	8.775	0.289	0.454
H4a	Perceived social value - > eWOM recommendation	0.091	**	2.659	0.024	0.159
H4b	Perceived ethical value - > eWOM recommendation	0.377	**	8.776	0.291	0.461
H4c	Perceived altruistic value - > eWOM recommendation	0.363	**	8.775	0.280	0.444
H5	Customer expectations x information overload - > greenwashing	0.111	**	3.164	0.041	0.179
H6a	Customer expectations x greenwashing - > social value	0.102	**	4.333	0.061	0.153
H6b	Customer expectations x greenwashing - > ethical value	0.070	**	2.787	0.026	0.125
H6c	Customer expectations x greenwashing - > altruistic value	0.100	**	4.091	0.024	0.159

**p < 0.01; *p < 0.05.

overload on greenwashing perception (H5; $\beta = 0.111$; $p < 0.01$) because of a significant moderation effect. Accordingly, the deterioration of the credibility of sustainability practices (greenwashing), and the increase in perception of information overload, are reinforced for those individuals who honestly believe in the need for hotels to perform sustainability practices.

Those results would not be a problem for firms if greenwashing did not have a significant negative impact on their perceived value, but it does, at least in most of its dimensions. Perceived greenwashing negatively affects customers' perceived value; it hurts perceived ethical-expressive value (H2b; $\beta = -0.087$; $p < 0.01$) and altruistic value (H2c; $\beta = -0.113$; $p < 0.01$), although the effect of greenwashing on social value (i.e., making a good impression to other people) is not significant (H2a; $\beta = -0.041$; $p > 0.05$).

Our results also prove that the negative effect of greenwashing on the perceived value of a hotel is weakened in those individuals with high expectations of what a hotel should do regarding sustainable measures, as those expectations positively moderate the effect of greenwashing on social value (H6a; $\beta = 0.102$; $p < 0.01$), ethical-expressive value (H6b; $\beta = 0.070$; $p < 0.01$) and altruistic value (H6c; $\beta = 0.100$; $p < 0.01$). These results seem to indicate that the more practices that a consumer expects that a hotel will undertake (i.e., higher customer expectations of hotel sustainability practices), the more likely it is that these customers accept some greenwashing, a kind of "make more claims about what you do than you really do but … at least do something".

Finally, the results indicate that everything that affects perceived value is an important question to be analyzed as the positive effects of perceived value dimensions on booking intentions (H3a; $\beta = 0.128$; $p < 0.01$; H3b; $\beta = 0.268$; $p < 0.01$ and H3c; $\beta = 0.363$; $p < 0.01$) and on eWOM recommendations (H4a; $\beta = 0.091$; $p < 0.01$; H4b; $\beta = 0.377$; $p < 0.01$ and H4c; $\beta = 0.363$; $p < 0.01$) are clearly significant.

5. Discussion

This research investigates the effectiveness of communicating sustainability practices by drawing from the S-O-R model, media richness theory and expectation-confirmation theory. Booking.com has transparently communicated a series of sustainability practices, self-reported by their suppliers, on the basis that i) customers with altruistic sustainability value orientations respond well to the detailed sustainability performance of hotels (Vinzenz et al., 2019) and ii) transparency improves customer trust and reduces skepticism (Bhaduri & Copeland, 2021). We show that perceived value of sustainability information positively impacts booking intentions and eWOM recommendations, but that such value depends on the customers' expectations and on their perceptions of greenwashing and of information as overload, as explained next.

First, our study contributes to understanding how much, and what type of, sustainability information helps customers to make booking decisions. We show that transparency is important but insufficient to reduce the perception of greenwashing. Companies must ensure that the information resonates with consumer expectations and provides some sort of experiential benefit. Few of the sustainability practices shown in Fig. 2 may be perceived as having a customer benefit (e.g., sourcing food locally, featuring local art, offering tours and activities from local guides, and bicycle rental). Using a sustainability scorecard to evaluate perceived information overload (stimuli in the S-O-R model), we suggest that "less is more". Selectively communicating aspects that enhance customer value is key, as consumers have limited attention spans and will perceive any information irrelevant to their immediate tasks as noise (Bawden & Robinson, 2020; Matthes et al., 2020). Hence, both online distribution channels and their suppliers need to learn how to translate the benefits of sustainability practices into enhanced service (Font *et al.*, 2021). Our findings reinforce recent evidence that sustainability messages must be congruent with customers' identities, and that

sustainability messages will be rejected when customers find green practices inconvenient (Tabatabaei & Beldona, 2024) or when they do not provide a positive customer experience (Hanna et al., 2018).

Second, drawing from the integration of the S-O-R model with the media richness theory, we examine the effect of perceived sustainability information overload on perceived greenwashing. Previous research suggests that communicating sustainability practices can help reduce cognitive dissonance and eco-paralysis (Li et al., 2024). However, the information being communicated needs to be perceived as trustworthy and helpful. In this study, about half of the respondents perceived Booking.com's sustainability information as excessive, unnecessary, and overly complex, resulting in a perception of information overload (Otondo et al., 2008; Bawden & Robinson, 2020; Matthes et al., 2020; Torma & Thøgersen, 2024). Many of the featured sustainability practices in Fig. 2 are now commonplace in many countries (recycling bins, room keycards, water efficient toilets/showers …) and would not impress customers. When booking a hotel, consumers prioritize task-related decisions. At this point in the customer journey, the social or emotional benefits of sustainability information are less salient and relevant (Otondo et al., 2008; Roetzel, 2019; Bawden & Robinson, 2020). The richness of sustainability information provided by Booking.com complicated customers' decision-making processes through information overload (Bogliacino et al., 2023; Longo et al., 2019; Weber, 2021). We show how the unintended negative consequences of information overload led to an increased perception of greenwashing (Rahman et al., 2015). As consumers often believe that hotels overstate their sustainability claims (Gupta et al., 2019; Peng & Chen, 2019), they may feel that they are exposed to information overload when the information is not substantiated by trustworthy claims (Gupta et al., 2019; Rahman et al., 2015). Our findings demonstrate that information overload leads to a perception that information is unclear and potentially misleading, generating skepticism about why such information was communicated and perceptions of greenwashing (Zhang et al., 2018). Hence, our study confirms recent findings that sustainability communication that is too frequent, ambiguous, or undifferentiated may increase customer confusion, which reduces its effectiveness (Torma & Thøgersen, 2024).

Third, we contribute to the expectation-confirmation theory (Olkkonen & Luoma-Aho, 2015; Zeithaml et al., 2018) by showing how customers' sustainability expectations act as standards against which a hotel's actual performance is evaluated. We offer insights into the moderating role of customer expectations of hotel sustainability practices on the influence of perceived information overload on perceived greenwashing. Our findings indicate that the effect of sustainability information overload (where customers are overwhelmed by the volume, or underwhelmed by its uselessness) on perceiving greenwashing is relative to customer expectations. Drawing from the examples in Fig. 2, we can see that some of the claims made are now legally required in many countries (e.g. provision of recycling bins or not using single use plastic amenities) and their communication is therefore redundant. To summarize, how customers perceive greenwashing is affected by both the amount and quality of the information they receive about sustainability practices and their expectations. Higher expectations of sustainability practices make communication claims easier to interpret, thus reducing the effect of information overload on greenwashing.

Fourth, based on the typology of customer value in the context of sustainability (Ahn & Kwon, 2020; Marchoo et al., 2014; Peloza & Shang, 2011), we focus on three dimensions of perceived value: social, ethical-expressive, and altruistic. Extrapolating from earlier work from Szabo and Webster (2021), we demonstrate that the perception that a hotel (and, by extension, Booking.com) engages in greenwashing decreases the perceived value of sustainability information. While customers may be willing to engage in some of the sustainability practices communicated in Fig. 2 for social, ethical-expressive, and altruistic reasons once on site (e.g. opting out of room cleaning, reusing towels), these are hardly going to be compelling reasons to book a property. Our findings show that promoting low effort sustainability practices is perceived as greenwashing, and these customers will only gain a sense of ethical-expressive and altruistic value from participating in higher level sustainability efforts. Based on the S-O-R model, perceived value is a psychological factor that affects consumers' responses (Hu et al., 2023) and, specifically, influences booking intentions and eWOM recommendations. We can expect that customers will speak highly of sustainability practices that are seen as meaningful or that stand out from those that they see regularly, but those featured in Fig. 2 do not convey that wow factor.

Our findings further reveal that when customers' expectations are low, the negative effect of greenwashing on perceived value is stronger, which is the opposite of what we expected (see H6). In other words, customers with low expectations are more negatively impacted by perceived greenwashing and, for these customers, the perceived value of hotels with sustainable measures is significantly reduced. However, when customer expectations are high, the negative effect of greenwashing on perceived value is less pronounced. This suggests that customers with high expectations may be more forgiving of, or less sensitive to, greenwashing, maintaining a higher perception of value despite greenwashing practices.

6. Conclusion

We tested the effects of perceived information overload of sustainability communication on customers' perceptions of greenwashing in online hotel bookings. We demonstrate that information overload increases the perception of greenwashing. In addition, we examined the moderating effect of customer expectations on the relationship between perceived information overload and perceived greenwashing. Customers with higher-than-average expectations of sustainability practices are less critical of sustainability claims, which leads to a positive moderating effect on the relationship between information overload and greenwashing. Our research findings also suggest that customers penalize hotels that they perceive as being dishonest, as demonstrated by the negative effect of greenwashing on ethical and altruistic values. However, despite the negative effect of greenwashing on perceived value, customers with high expectations of sustainability practices perceive greater social, ethical-expressive, and altruistic value from staying at green hotels. Finally, perceived value positively impacts both booking intentions and eWOM recommendations.

The lesson learned with managerial implications is that companies must scan their operating environment to comply with legislation, best illustrated by the now defunct Booking.com Travel Sustainable badge, which highlighted the self-reported sustainability practices of over 400,000 properties. The logic behind displaying such practices was to encourage customers to hold the properties accountable via customer reviews. The Travel Sustainable Badge was removed in March 2024 due to pressure from the Netherlands Authority for Consumers and Markets with reference to the European Green Claims Directive (European Parliament, 2024), and what has remained (as of January 2025) is the property-level information on each hotel's sustainability practices. Booking.com, 2022 now presents this information as self-reported, toning down its sustainability claims. However, now that these sustainability claims are less visible in the Booking platform, the accommodation businesses no longer have the same incentive to report their sustainability practices. Hence, a risk analysis is needed to understand the negative, unintended consequences of sustainability communication campaigns. Companies ought to communicate less sustainability information that is trustworthy and verified (Majeed & Kim, 2023).

Finally, we reflect on some limitations of our study that can lead to further research avenues. First, research could expand on the meaning of, and reasons behind, perceived information overload by investigating the number and the types of sustainability practices communicated. Future research could examine whether sustainability practices are perceived as a significant effort by the hotel and whether they enhance

customer experience (Font et al., 2021). Second, while our study focuses on three dimensions of customer value, future research should examine the impact of different types of values based on the self-oriented/other-oriented and extrinsic/intrinsic dimensions, including the use of customer rewards for sustainable choices (Peloza & Shang, 2011; Vinzenz et al., 2019). Third, future research could delve into how digital communication methods, such as information filtering, can decrease information overload (Li, 2017). Finally, there are methodological weaknesses from a survey of self-reported attitudes and behavioral intentions (Viglia et al., 2024), that could be addressed in two ways: i) eye-tracking and neuroscience tools would enable identifying the visual attention of sustainability cues that customers gravitate towards (Li et al., 2024), and ii) field experiments with an online travel agent would help us to understand the actual changes in booking patterns affected by specific sustainability information communications, and how that information could be best framed (Liu et al., 2024).

CRediT authorship contribution statement

Xavier Font: Writing – original draft, Supervision, Investigation, Conceptualization. **Luisa Andreu:** Writing – original draft, Project administration, Methodology, Investigation. **Anna S. Mattila:** Writing – original draft, Validation. **Joaquín Aldas-Manzano:** Writing – original draft, Methodology, Formal analysis, Data curation.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jhtm.2025.01.013.

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